

Nondeterministic Bottom-Up Parsing

Recap: Leftmost analysis with NTA

- We have seen the nondeterministic Top-Down parsing automaton that gives the leftmost analysis of an input

- Example

$E \rightarrow E + T \mid T$	$(1, 2)$
$T \rightarrow T * F \mid F$	$(3, 4)$
$F \rightarrow (E) \mid a \mid b$	$(5, 6, 7)$

on input $(a) * b$

- | | |
|------------------------|---|
| • Initial state | $((a) * b, E, \varepsilon)$ |
| • Rule 2 to expand E | $((a) * b, T, 2)$ |
| • Rule 3 to expand T | $((a) * b, T * F, 23)$ |
| • Rule 4 to expand T | $((a) * b, F * F, 234)$ |
| • Rule 5 to expand E | $((a) * b, (E) * F, 2345)$ |
| • Match the "(" | $(a) * b, E) * F, 2345)$ |
| • Rule 2 to expand E | $(a) * b, T) * F, 23452)$ |
| • Rule 4 to expand T | $(a) * b, F) * F, 234524)$ |
| • Rule 6 to expand F | $(a) * b, a) * F, 2345246)$ |
| • Match "a" | $() * b,) * F, 2345246)$ |
| • Match ")" | $(* b, * F, 2345246)$ |
| • Match "*" | $(b, F, 2345246)$ |
| • Rule 7 to expand F | $(b, b, 23452467)$ |
| • Match "b" | $(\varepsilon, \varepsilon, 23452467)$ |

Bottom-up parsing

- Idea: Parse from the leafs of the syntax tree toward the root

- Example
$$\begin{array}{ll} E \rightarrow E + T \mid T & (1, 2) \\ T \rightarrow T * F \mid F & (3, 4) \\ F \rightarrow (E) \mid a \mid b & (5, 6, 7) \end{array}$$

on input $(a) * b$

- (video)

Nondeterministic Bottom-Up Automaton (NBA)

- Similar to NTA, we define the NBA for $G = \langle \Sigma, N, P, S \rangle$ by
 - Input alphabet Σ
 - Pushdown alphabet $X = N \cup \Sigma$
 - Output alphabet $U =$ the rule numbers $1, 2, 3, \dots$
 - States $\Sigma^* \times X^* \times U^*$
 - Two types of transitions for $w \in \Sigma^*, \alpha \in X, z \in U$:
 - **Reducing** using rule $A \rightarrow \beta$ with number i :
$$(w, \alpha\beta, z) \rightarrow (w, \alpha A, z i)$$
 - **Shifting** of terminal symbol $a \in \Sigma$:
$$(aw, \alpha, z) \rightarrow (w, \alpha a, z)$$
 - Initial state $(w, \varepsilon, \varepsilon)$ for $w \in \Sigma^*$
 - Final state (ε, S, u) where $u \in U^*$
- **Theorem (without proof):** Running the NBA on input w gives the reversed rightmost analysis u of w

Example: Reversed Rightmost Analysis with NBA

$$\begin{aligned} E &\rightarrow E + T \mid T & (1, 2) \\ T &\rightarrow T * F \mid F & (3, 4) \\ F &\rightarrow (E) \mid a \mid b & (5, 6, 7) \end{aligned}$$

■ Running the NBA on $w = (a) * b$

- Initial state $((a) * b, \quad \varepsilon, \varepsilon)$
- Shift $(\ a) * b, \quad (, \varepsilon)$
- Shift $(\) * b, \quad (a, \varepsilon)$
- Rule 6 to reduce a to F $(\) * b, \quad (F, 6)$
- Rule 4 to reduce F to T $(\) * b, \quad (T, 64)$
- Rule 2 to reduce T to E $(\) * b, \quad (E, 642)$
- Shift $(\ * b, \quad (E), 642)$
- Rule 5 to reduce (E) to F $(\ * b, \quad F, 6425)$
- Rule 4 to reduce F to T $(\ * b, \quad T, 64254)$
- Shift $(\ b, \quad T *, 64254)$
- Shift $(\ \varepsilon, \quad T * b, 64254)$
- Rule 7 to reduce b to F $(\ \varepsilon, \quad T * F, 642547)$
- Rule 3 to reduce $T * F$ to T $(\ \varepsilon, \quad T, 6425473)$
- Rule 2 to reduce T to E $(\ \varepsilon, \quad E, 64254732)$

Nondeterminism

- Like NTAs, NBA parsing can be nondeterministic:
 - If we have a rule $A \rightarrow a$ and the state $(bw, \alpha a, z)$, should we reduce a to A or shift b ?
 - If we have rules $A \rightarrow ab$ and $B \rightarrow b$ and the state $(w, \alpha ab, z)$, should we reduce ab to A or b to B ?
 - If we have rules $A \rightarrow a$ and $B \rightarrow a$ and the state $(w, \alpha a, z)$, should we reduce a to A or to B ?
 - If we have rule $A \rightarrow S$ (where S is the initial symbol) and the state (ε, S, z) , should we stop parsing or reduce S to A ?
- The last case can be easily fixed:

For grammars like

$$\begin{aligned} S &\rightarrow \dots \\ A &\rightarrow S \end{aligned}$$

we add a new initial symbol S' and rule

$$S' \rightarrow S$$

This is called a *start-separated* grammar. No further reduction possible after reaching (ε, S', z)